SE Lab 1 – Requirements Engineering & UML Use-Case Modelling

Problem Statement #04: Student Project Portfolio & Showcase App

Campus & Academic Operations

Name: Heemadhawala R

SRN:: PES2UG24CS190

SEM:5

Section:D

Problem Statement #04 – Student Project Portfolio & Showcase App

1. Requirements Table

The Lab 1 handout requires exactly five Functional Requirements and two Non-Functional Requirements, each with a measurable acceptance criterion and rationale. The problem statement supplies FR-001 and NFR-001 as guidelines; they are retained/refined below.

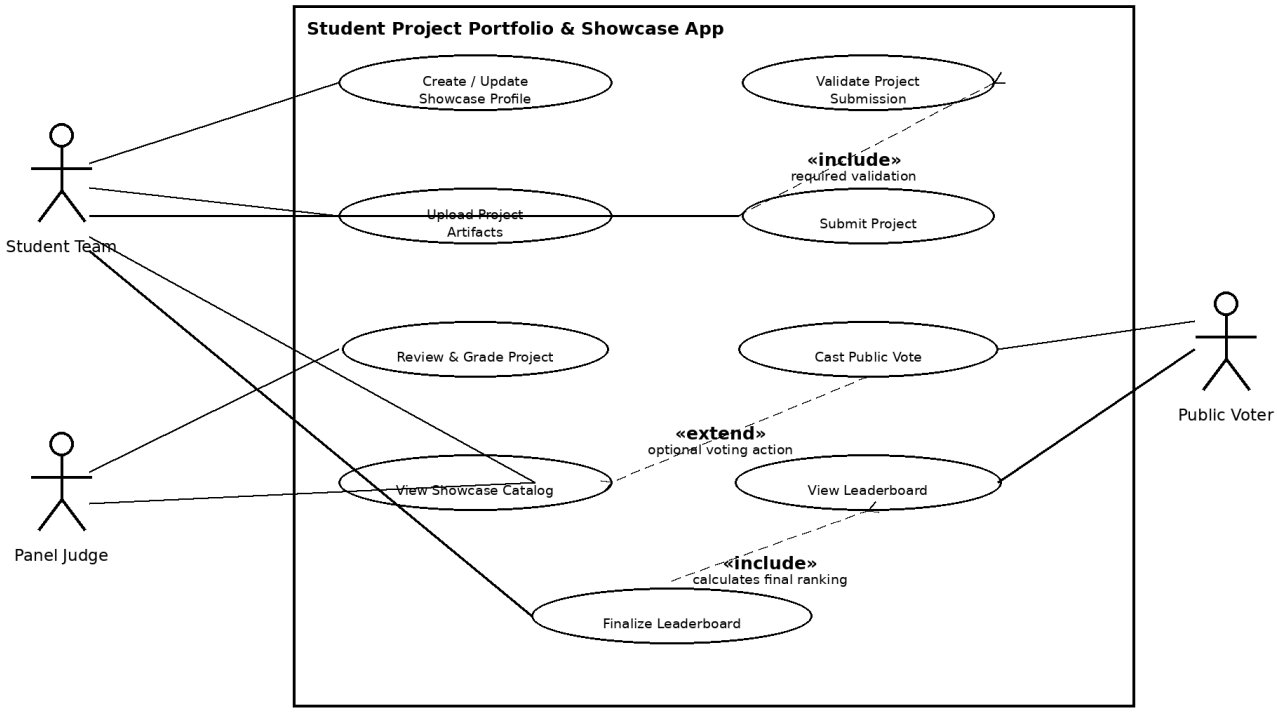
Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall enable student teams to create showcase profiles containing abstract summaries, demo video links, and verified GitHub repository URLs.	High	Pass: A profile is published to the showcase catalog only after the repository URL passes validation and all mandatory profile fields are present. Fail: An invalid repository URL or incomplete profile is published.	Provides the core mechanism for student teams to present their projects in the annual showcase.	Given in problem statement; refined with measurable validation.
FR-002	Functional	The system shall allow student teams to upload and attach approved multimedia project artifacts, including images, presentation files, and demonstration videos, to their showcase profile.	High	Pass: Each supported artifact uploads successfully, is linked to the correct project profile, and is visible in the team's artifact list. Fail: An unsupported or corrupted file is accepted as a project artifact.	Allows teams to provide evidence and demonstrations of their project work.	Supports multimedia showcase requirements.
FR-003	Functional	The system shall allow a panel judge to access submitted project profiles and grade them using a predefined digital rubric, with each criterion receiving a score and the system calculating the project's total score.	High	Pass: Every required rubric criterion receives a valid score and the system calculates the total according to the configured rubric weights. Fail: A submission can be finalized with a missing criterion score or an incorrect total.	Digitizes and standardizes project evaluation by panel judges.	Ensures consistent and traceable judging.
FR-004	Functional	The system shall allow verified public voters to cast at most one vote per project during the active voting period and shall update the public leaderboard using the recorded votes.	High	Pass: A verified voter can submit one vote for a project, while a second vote for the same project is rejected and recorded as a duplicate attempt. The leaderboard reflects accepted votes.	Provides the public voting mechanism while preventing repeated votes for the same project.	Implements public participation with a basic per-project anti-sybil control.
FR-005	Functional	The system shall publish the final showcase leaderboard after the configured voting deadline and make each project's final judge score, public-vote count, and overall ranking visible according to the showcase publication rules.	High	Pass: Before the deadline the final leaderboard is not locked; after the deadline the ranking is finalized and each published project displays its permitted score, vote count, and rank. Fail: Rankings remain editable after finalization or required published fields are missing.	Provides a transparent final result for the annual project showcase.	Completes the showcase and result-publication workflow.
NFR-001	Nonfunctional – Performance & Security	The public voting and leaderboard calculation services shall maintain	High	Pass: Load testing shows at least 95% of voting/leaderboard API	Matches the problem statement's real-time performance and	Given in problem statement; expanded

		an API response latency below 300 ms for at least 95% of requests under the defined simulated peak-load test, while applying the configured anti-sybil verification controls.		requests complete in under 300 ms and repeated/invalid voting attempts are rejected by the configured verification controls. Fail: The latency target or voting-control requirement is not met.	security expectation for public voting and leaderboard calculations.	into a measurable test condition.
NFR-002	Nonfunctional – Availability & Data Integrity	The system shall preserve submitted project artifacts, judge scores, accepted votes, and finalized leaderboard results with no data loss during normal operation and shall recover from a service interruption without duplicating accepted votes.	High	Pass: After a simulated restart/recovery, all previously committed records remain available and each accepted vote appears exactly once. Fail: Any committed record is lost, corrupted, or duplicated.	Protects project submissions and evaluation results from data loss and preserves trust in the showcase.	Important for a high-stakes annual showcase and public ranking.

2. UML Use-Case Diagram

Actors are derived from the scenario: Student Team and Panel Judge are explicitly listed as target actors, while Public Voter is added because the scenario explicitly states that the public participates in voting. The lab handout requires at least three actors and at least one «include» and one «extend» relationship.

UML Use-Case Diagram – Student Project Portfolio & Showcase App



Actors: Student Team, Panel Judge, Public Voter | Required relationships: «include» and «extend»

«include» is used for mandatory reusable behavior such as submission validation; «extend» is used for optional/conditional behavior such as casting a public vote from the showcase catalog.

3. Use-Case Flow Specification

Use Case: Create / Submit Student Project Showcase Profile

Preconditions

- The Student Team is authenticated and has permission to create or edit its project showcase profile.
- The project showcase period is open for submissions.
- The team has a project abstract, demo video link, and GitHub repository URL ready for submission.
- The supplied GitHub repository URL is reachable for validation.

Postconditions

- A showcase profile is created or updated for the student project.
- The abstract, demo video link, and validated GitHub repository URL are stored with the project.
- The project is marked as Submitted/Published according to the validation and showcase rules.
- The project becomes available in the showcase catalog for eligible judges and, when the voting period is active, public voters.

Main Success Scenario

1. Student Team selects “Create/Update Project Showcase Profile”.

2. System displays the project profile form.

3. Student Team enters the project abstract and demo video link and provides the GitHub repository URL.

4. System validates required fields and checks the GitHub repository URL.

5. System validates the demo video link format and required profile information.

6. Student Team uploads or attaches the required multimedia project artifacts.

7. System stores the profile and artifacts and generates a unique project entry.

8. System marks the project as ready for showcase submission/publishing.

9. System displays a confirmation containing the project entry ID and current submission status.

10. Use case ends successfully.

Alternate Flow

4a. Invalid GitHub repository URL: The system rejects the submission, identifies the invalid URL, and asks the Student Team to provide a valid repository URL.

4a1. Student Team corrects the URL and resubmits it.

4a2. The system revalidates the URL and resumes at Step 5.

6a. Unsupported or corrupted multimedia artifact: The system rejects the file, displays the supported file requirements, and allows the Student Team to upload a valid artifact or continue with the remaining required items.

Actors Involved

Primary Actor: Student Team

Supporting Actors: Panel Judge, Public Voter